Contents

1 Continuous Functions

1.1 Continuity of a Function

Definition 1.1.1

Let X and Y be topological spaces. A function $f : X \to Y$ is said to be ***continuous*** if for each open subset V of Y, the set $f^{-1}(V)$ is an open subset of X.

Theorem 1.1.2

Let X and Y be topological spaces; let $f : X \to Y$. Then the following are equivalent:

1. f is continuous.
2. For every subset A of X, one has $f(\bar{A}) \subset \overline{f(A)}$.
3. For every closed set B of Y, the set $f^{-1}(B)$ is closed in X.
4. For each $x \in X$ and each neighborhood V of $f(x)$, there is a neighborhood U of x such that $f(U) \subset V$.

Remark.
If the condition in (4) holds for the point x of X, we say that f is ***continuous at the point x***.

Proof. 1. $\implies$ 2. : If f is continuous, take any $x \in f(\bar{A})$. Let V be any neighborhood of $f(x)$. Then $f^{-1}(V)$ is a neighborhood of x. We have $f^{-1}(V) \cap A \neq \varnothing$. Then

$$\varnothing \neq f\left(f^{-1}(V) \cap A\right) \subseteq V \cap f(A)$$

Since V is arbitrary, $f(x) \in \overline{f(A)}$. □

1.2 Homeomorphisms

Definition 1.2.1

Let X and Y be topological spaces; let $f : X \to Y$ be a bijection. If both the function f and the inverse function

$$f^{-1} : Y \to X$$

are continuous, then f is called a ***homeomorphism***.